Notes 01.03: Course Schedule, Pedagogy & Roadmap

1. The 60-Day Execution Philosophy

Engineering proficiency in AI/ML cannot be achieved through passive video observation. The curriculum follows the **Active Implementation Pedagogy**:

$$\text{Mastery} = \text{Theory Derivation (First Principles)} + \text{Hands-on Code} + \text{Debugging}$$

The Daily 3-Step Routine:

1. **Understand the Math:** Why does the equation work? What is the physical intuition?
2. **Implement in Code:** Write clean, modular Python/NumPy code before relying on high-level Scikit-Learn abstractions.
3. **Analyze Edge Cases:** Test on noisy data, measure complexity, tune hyperparameters, and visualize outcomes.

---

2. Project Milestones & Portfolio Capstones

A stellar GitHub portfolio requires tangible, production-tested artifacts:

- **Milestone 1: Web Scraping & Automated Data Pipeline (Day 17)**
  End-to-end data pipeline scraping unstructured web pages, cleaning tables with Pandas, and writing to SQL.
- **Milestone 2: CreditWise Loan Underwriting System (Day 27 — Minor Project)**
  Supervised classification pipeline predicting loan default risk with cost-sensitive threshold optimization.
- **Milestone 3: SmartCart Customer Clustering System (Day 33 — Minor Project)**
  Unsupervised segmentation clustering e-commerce shoppers based on RFM (Recency, Frequency, Monetary) metrics.
- **Milestone 4: Text Summarizer with Modern NLP (Day 48 — Minor Project)**
  Sequence-to-sequence summarization using fine-tuned Hugging Face Transformer models.

- **Milestone 5: Production Capstone with Full MLOps CI/CD (Days 56–60)**
  Containerized full-stack ML application with Flask/FastAPI backend, automated unit tests, and cloud deployment.